

BAND-ENERGY ADAPTIVE GRAPH FILTERING FOR ENSEMBLE CLUSTERING

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ABSTRACT

Ensemble clustering aggregates base partitions into a consensus, but the regularization applied to the ensemble-induced graph is usually fixed. We propose band-energy adaptive graph filtering (BEGF), which constructs a normalized co-membership kernel directly from base partitions and uses the fixed Laplacian $L = 2(I - X_0 X_0^\top)$. Its spectrum lies in $[0, 2]$, and matrix-vector products are evaluated without forming an $n \times n$ graph. A degree-two Bernstein partition resolves Laplacian variation into low-, middle-, and high-frequency bands. A convex pseudo-Huber penalty maps normalized band energies to bounded MM weights, inducing a signal-dependent spectral response while retaining a strongly convex objective and SPD linear updates. On ten benchmarks with 165 to 149,332 samples, BEGF is best or tied-best in 14 of 30 dataset-metric cells. On the three datasets above 130,000 samples, it is best in five of nine cells and second-best in the remaining four.

Index Terms— ensemble clustering, graph signal processing, graph filtering, M-estimation, spectral regularization

1. INTRODUCTION

Ensemble clustering (EC) combines multiple base partitions into a single consensus without requiring the original features. Classical consensus functions use co-association, hypergraph partitioning, or related evidence aggregation [1, 2, 3, 4]. Recent methods improve finite-ensemble generalization, co-association construction, topological kernel learning, and large-scale robust aggregation [5, 6, 7, 8]. Across these lines, the main design variables are the evidence entering consensus, its weighting, graph construction, or the representation learned from that evidence.

The regularizer acting on a fixed ensemble-induced graph is a separate design variable. Once a global quadratic smoothing level is chosen, the relative attenuation assigned to different Laplacian frequencies is predetermined. Yet the filtered EC signal encodes normalized memberships of base clusters: agreement repeatedly supported by the ensemble and boundary ambiguity can appear at different spectral scales. Their realized magnitudes can therefore inform how strongly each spectral region should be regularized. Graph signal processing provides the spectral language for this view [9], while Bernstein bases provide a nonnegative partition of the Laplacian spectrum [10]. Smooth pseudo-Huber penalties provide a convex way to reduce marginal shrinkage on unusually energetic components [11].

We develop band-energy adaptive graph filtering (BEGF). The method (i) constructs a fixed Laplacian directly from the normalized

ensemble co-membership kernel, (ii) resolves its Dirichlet variation into three Bernstein bands, and (iii) uses a convex bandwise M-type objective whose MM weights are explicit functions of current normalized band energies. The resulting solver needs only low-order applications of the fixed Laplacian and conjugate gradients (CG). Ordinary quadratic graph filtering is recovered exactly when adaptation is disabled.

2. BAND-ENERGY ADAPTIVE GRAPH FILTERING

2.1. Ensemble representation and fixed Laplacian

Let m base partitions be given. Partition r has c_r nonempty clusters and indicator matrix $H^{(r)} \in \{0, 1\}^{n \times c_r}$ with one nonzero per row. Define

$$D^{(r)} = H^{(r)\top} H^{(r)}, \quad F^{(r)} = H^{(r)} D^{(r)-1/2}, \quad (1)$$

and concatenate the normalized indicators as

$$X_0 = \frac{1}{\sqrt{m}} [F^{(1)}, \dots, F^{(m)}] \in \mathbb{R}^{n \times q}, \quad q = \sum_{r=1}^m c_r. \quad (2)$$

Each base-cluster indicator has unit norm. Moreover,

$$K_0 := X_0 X_0^\top = \frac{1}{m} \sum_{r=1}^m H^{(r)} D^{(r)-1} H^{(r)\top} = \frac{1}{m} \sum_{r=1}^m F^{(r)} F^{(r)\top}. \quad (3)$$

Thus K_0 is a normalized co-membership kernel: a base cluster contributes pairwise evidence inversely proportional to its size. Since each $F^{(r)} F^{(r)\top}$ is an orthogonal projector, $0 \preceq K_0 \preceq I_n$. We use the fixed ensemble-induced Laplacian

$$L = 2(I_n - K_0) = 2(I_n - X_0 X_0^\top). \quad (4)$$

It follows that $0 \preceq L \preceq 2I_n$; also $K_0 \mathbf{1} = \mathbf{1}$, hence $L \mathbf{1} = \mathbf{0}$. The operator is never formed explicitly. For any block Z ,

$$LZ = 2[Z - X_0 (X_0^\top Z)]. \quad (5)$$

The variable $Y \in \mathbb{R}^{n \times q}$ denotes the filtered continuous ensemble representation.

2.2. Frequency-resolved variation and objective

Write $L = U \Lambda U^\top$, where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ and $\lambda_i \in [0, 2]$. For $u = \lambda/2$, define

$$\phi_{\text{lo}} = (1 - u)^2, \quad \phi_{\text{mid}} = 2u(1 - u), \quad \phi_{\text{hi}} = u^2, \quad (6)$$

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so that $\sum_{b \in \mathcal{B}} \phi_b(\lambda) = 1$ for $\mathcal{B} = \{\text{lo}, \text{mid}, \text{hi}\}$. With $\hat{Y} = U^\top Y$, define the band energy

$$E_b(Y) = \sum_{i=1}^n \lambda_i \phi_b(\lambda_i) \|\hat{y}_i\|_2^2, \quad \sum_b E_b(Y) = \text{tr}(Y^\top LY). \quad (7)$$

Let $s_b = E_b(X_0) + \varepsilon$ with $\varepsilon > 0$. For $z \geq 0$, use

$$\rho_{\delta, \eta}(z) = \frac{\eta}{2} z^2 + (1 - \eta) \delta^2 \left(\sqrt{1 + (z/\delta)^2} - 1 \right), \quad (8)$$

where $\delta > 0$ and $0 \leq \eta \leq 1$. BEGF solves

$$Y^* = \arg \min_Y \frac{1}{2} \|Y - X_0\|_F^2 + \mu \sum_{b \in \mathcal{B}} s_b \rho_{\delta, \eta} \left(\sqrt{E_b(Y)/s_b} \right). \quad (9)$$

The band term is convex because $\sqrt{E_b(Y)/s_b}$ is a seminorm and $\rho_{\delta, \eta}$ is convex and nondecreasing on $[0, \infty)$. The fidelity term is 1-strongly convex, so (9) has a unique global minimizer.

2.3. MM solver and induced response

Define $R_b = U \text{diag}(\lambda_i \phi_b(\lambda_i)) U^\top$. From (6),

$$R_{\text{lo}} = L - L^2 + \frac{1}{4} L^3, \quad R_{\text{mid}} = L^2 - \frac{1}{2} L^3, \quad R_{\text{hi}} = \frac{1}{4} L^3, \quad (10)$$

with $R_b \succeq 0$ and $\sum_b R_b = L$. Majorizing the concave square-root term at $E_b^{(t)} = E_b(Y^{(t)})$ yields

$$w_b^{(t)} = \eta + \frac{1 - \eta}{\sqrt{1 + E_b^{(t)}/(s_b \delta^2)}} \in [\eta, 1]. \quad (11)$$

Hence larger normalized energy gives smaller next-step quadratic shrinkage. The MM update is the SPD system

$$\left[I_n + \mu \sum_b w_b^{(t)} R_b \right] Y^{(t+1)} = X_0. \quad (12)$$

CG solves (12) using only repeated applications of (5); neither U , K_0 , L , nor their powers are stored. At frequency λ the induced response is

$$g_t(\lambda) = [1 + \mu \lambda a_t(\lambda)]^{-1}, \quad a_t(\lambda) = \sum_b w_b^{(t)} \phi_b(\lambda), \quad (13)$$

with $(1 + \mu \lambda)^{-1} \leq g_t(\lambda) \leq (1 + \mu \eta \lambda)^{-1}$. If $\eta = 1$, all weights equal one and (12) reduces exactly to $(I_n + \mu L)Y = X_0$. Unequal weights reshape the spectrum; nearly equal weights reduce BEGF to a globally rescaled smoother.

3. EXPERIMENTS

3.1. Setup

We evaluate on ten datasets summarized in Table 1. All methods receive the same frozen base-partition input for a dataset. We compare with CSPA, HGPA, and MCLA [1], EAC [2], KCC [3], ECPCS-HC [4], GPEC [5], SDGCA [6], OMELET [7], and RANGE [8]. Performance is measured by ACC, ARI, and NMI. Missing valid outputs are not imputed and are shown as “–”.

Table 1. Characteristics of all datasets.

ID	Dataset	#Sample	#Feature	#Class
D1	Yale	165	14,150	15
D2	ProteinFold	694	324	27
D3	COIL20	1,440	1,024	20
D4	Communities	1,994	106	3
D5	Obesity	2,111	31	7
D6	USPS	9,298	256	10
D7	PAMAP2	18,716	128	7
D8	EMNIST-Balanced	131,600	784	47
D9	EMNIST-Letters	145,600	784	26
D10	Walking	149,332	4	22

Table 2. Clustering performance on all ten datasets, measured by ACC, ARI, and NMI (%). The best results among ensemble clustering methods are in bold. Base clusterings (avg.) reports the arithmetic mean performance of the frozen base partitions. “–” denotes unavailable valid results.

Method	Yale			ProteinFold			COIL20			Communities			Obesity		
	ACC	ARI	NMI	ACC	ARI	NMI	ACC	ARI	NMI	ACC	ARI	NMI	ACC	ARI	NMI
Base clusterings (avg.)	64.06	43.31	65.63	21.54	6.92	29.63	65.08	60.13	78.48	51.48	15.10	15.08	42.67	25.05	31.22
CSPA [1]	37.58	16.10	40.19	30.69	13.78	40.03	70.00	63.82	78.96	50.05	12.88	13.05	36.62	18.02	25.03
HGPA [1]	48.48	29.03	54.47	24.64	8.60	29.50	50.35	40.92	65.38	44.38	5.74	6.01	34.72	14.85	19.25
MCLA [1]	58.18	35.50	58.71	32.56	14.90	40.07	71.60	65.41	80.58	51.50	15.13	15.10	40.93	22.03	27.52
EAC [2]	52.12	31.60	58.24	13.11	-0.08	11.81	64.31	43.38	81.63	51.50	15.13	15.10	42.35	20.86	27.59
KCC [3]	54.55	32.43	57.10	30.12	14.31	37.11	71.32	59.40	78.92	–	–	–	43.82	22.69	29.52
ECPCS-HC [4]	55.15	35.35	59.44	31.41	3.83	38.76	65.90	62.81	80.65	51.50	15.13	15.10	39.18	20.48	28.91
GPEC [5]	60.61	36.69	59.93	35.30	20.08	45.36	68.26	61.64	79.27	51.50	15.13	15.10	43.15	25.40	31.60
SDGCA [6]	39.39	25.66	51.26	32.52	15.03	37.27	68.19	63.40	81.39	45.89	4.88	4.51	41.10	17.97	26.74
OMELET [7]	60.00	34.08	58.84	34.87	16.29	43.20	70.63	66.61	81.73	51.50	15.13	15.10	43.25	25.07	30.87
RANGE [8]	55.15	32.97	57.60	33.43	17.09	42.06	48.40	40.83	72.45	51.50	15.13	15.10	41.97	22.94	28.38
BEGF (Ours)	69.09	49.94	71.69	38.18	19.07	44.22	70.76	66.69	81.72	51.50	15.13	15.10	42.59	24.94	30.60

Method	USPS			PAMAP2			EMNIST-Bal.			EMNIST-Lett.			Walking		
	ACC	ARI	NMI	ACC	ARI	NMI	ACC	ARI	NMI	ACC	ARI	NMI	ACC	ARI	NMI
Base clusterings (avg.)	63.46	52.51	59.34	50.56	34.16	45.28	27.58	14.31	37.15	35.69	19.46	37.18	22.28	9.59	25.91
CSPA [1]	56.60	41.98	51.47	69.25	51.99	58.52	–	–	–	–	–	–	–	–	–
HGPA [1]	11.41	0.00	0.18	46.42	26.27	39.05	20.57	9.74	28.07	24.41	12.31	26.04	5.51	-0.01	0.00
MCLA [1]	63.43	52.48	59.32	64.56	48.62	57.81	27.56	14.13	36.79	35.71	19.38	37.01	21.66	9.32	25.94
EAC [2]	63.45	52.51	59.35	17.31	0.00	0.08	–	–	–	–	–	–	–	–	–
KCC [3]	58.39	48.30	56.68	72.01	56.50	62.06	25.10	12.00	34.74	35.82	19.16	36.86	21.54	8.34	23.72
ECPCS-HC [4]	63.45	52.51	59.35	52.38	34.85	57.23	–	–	–	–	–	–	–	–	–
GPEC [5]	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
SDGCA [6]	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
OMELET [7]	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
RANGE [8]	63.45	52.51	59.35	74.78	61.36	66.60	26.76	14.16	36.56	35.76	19.35	36.99	22.89	9.88	25.65
BEGF (Ours)	63.43	52.48	59.31	69.73	58.58	67.71	27.17	14.28	37.06	35.79	19.57	37.07	22.47	9.58	26.00

Table 3. Ablation study of the main components in BEGF on all ten datasets. Results are measured by ACC, ARI, and NMI (%). Best displayed values are in bold.

Variant	Components		D1			D2			D3			D4			D5		
	Bands	Adapt.	ACC	ARI	NMI	ACC	ARI	NMI	ACC	ARI	NMI	ACC	ARI	NMI	ACC	ARI	NMI
w/o Bands	✓		69.09	49.94	71.69	38.18	19.07	44.22	70.69	66.65	81.60	51.50	15.13	15.10	42.59	24.94	30.60
w/o Adapt.	✓		69.09	49.94	71.69	38.04	19.00	44.24	70.76	66.67	81.61	51.50	15.13	15.10	42.59	24.94	30.60
BEGF	✓	✓	69.09	49.94	71.69	38.18	19.07	44.22	70.76	66.69	81.72	51.50	15.13	15.10	42.59	24.94	30.60

Variant	Components		D6			D7			D8			D9			D10		
	Bands	Adapt.	ACC	ARI	NMI	ACC	ARI	NMI	ACC	ARI	NMI	ACC	ARI	NMI	ACC	ARI	NMI
w/o Bands	✓		63.43	52.48	59.31	69.12	58.29	67.51	27.25	14.36	37.09	35.84	19.55	37.05	22.47	9.58	26.00
w/o Adapt.	✓		63.43	52.48	59.31	69.12	58.29	67.51	27.23	14.32	37.04	35.87	19.57	37.12	22.47	9.58	26.00
BEGF	✓	✓	63.43	52.48	59.31	69.73	58.58	67.71	27.17	14.28	37.06	35.79	19.57	37.07	22.47	9.58	26.00

3.2. Overall performance

Table 2 shows that BEGF is best or tied-best in 14 of 30 dataset-metric cells and second-best in 10 more. The strongest gains occur on Yale, where BEGF leads all three metrics, and it also attains the best ACC on ProteinFold and the best ARI on COIL20. The Communities data produce broad ties, while Obesity and PAMAP2 show that the proposed regularizer need not dominate every consensus construction. This is consistent with BEGF changing the spectral penalty on fixed ensemble evidence rather than replacing the evidence itself.

The large-scale cases are especially informative. EMNIST-Balanced, EMNIST-Letters, and Walking contain 131,600–149,332 samples. Across their nine reported metrics, BEGF is best in five and second-best in the remaining four. In particular, it leads ARI and NMI on both EMNIST variants and NMI on Walking. These cases also exercise the matrix-free operator in (5), avoiding explicit sample-by-sample graph storage in the filtering stage.

3.3. Ablation and implementation

Table 3 isolates the two mechanism components. “w/o Bands” replaces the three band energies by one global M-type penalty, while “w/o Adapt.” retains the three-band decomposition but removes energy-dependent reweighting. Across all ten datasets, BEGF averages 49.071 ACC, 33.026 ARI, and 47.048 NMI. The corresponding values are 49.016, 32.999, and 47.017 for w/o Bands, and 49.010, 32.992, and 47.022 for w/o Adapt. The average gains are modest because several datasets are nearly insensitive to spectral reshaping,

whereas the clearest improvements appear on PAMAP2 and COIL20. This agrees with (13): if the three band weights are nearly equal, the adaptive profile approaches a globally rescaled Laplacian smoother.

The parameters have separate roles. μ controls overall graph regularization, δ controls the transition away from the locally quadratic pseudo-Huber regime, and η lower-bounds each band weight; $\eta = 1$ removes adaptation exactly. Combining (10) gives

$$\sum_b w_b R_b = w_{lo} L + (w_{mid} - w_{lo}) L^2 + \frac{1}{4} (w_{lo} - 2w_{mid} + w_{hi}) L^3, \quad (14)$$

so each CG product uses at most three successive applications of the matrix-free operator in (5).

4. CONCLUSION

We presented BEGF, a fixed-graph regularization mechanism for ensemble clustering in which normalized band energies determine the spectral shrinkage applied at the next MM step. The revised ensemble-induced Laplacian $L = 2(I - X_0 X_0^\top)$ is symmetric PSD, has spectrum in $[0, 2]$, and supports direct matrix-free products from the normalized base-partition representation. The three-band construction preserves total Dirichlet variation, while the convex M-type objective yields bounded energy-dependent weights, SPD updates, and exact recovery of ordinary quadratic filtering. Experiments on ten datasets, including three with more than 130,000 samples, show competitive overall accuracy and particularly consistent large-scale performance.

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